

PAPER_018: Aether Noise Spectrum Characterization for LISA

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Domain: 1.2 $\hat{\alpha}$ Gravitational Waves $\hat{\alpha}$ LISA Future Detector

Primary Validation File: `validate_lisa_extended.py`

Calibration constants: $\hat{I}^0 = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$

Abstract

We characterize the spectral signature of UQFF aether fields in the LISA frequency band (0.1–100 mHz). UQFF predicts that U_m vacuum field oscillations imprint narrow spectral lines at harmonics of $f_U \hat{\alpha} 1$ mHz onto the stochastic gravitational wave background (SGWB), with an integrated aether power fraction of 222.93% relative to the GR SGWB. The peak aether feature at $f_{\text{peak}} = 0.99$ mHz yields an integrated detection SNR of 12,695,834 $\hat{\alpha}$ trivially detectable by LISA over a 4-year mission. A broad TRZ suppression dip ($\sim 10\%$) appears near 5 mHz. These spectral features have no astrophysical analogue and constitute a smoking-gun test of UQFF vacuum structure.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^4 \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis $\hat{\alpha}$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Motivation

UQFF predicts that the vacuum supports actively coupled aether fields characterized by: - **U_m** : Magnetic energy parameter ($= 1.0$ in calibrated UQFF) - **$\hat{I}^2_m$** : Modulation parameter ($= 0.01$) - **f_{TRZ}** : Trans-zero frequency factor ($= 0.1$)

These fields interact with propagating GWs to create: 1. Spectral **power excess** at fundamental and harmonic frequencies 2. **Sideband pairs** from $\hat{I}^2_m$ modulation 3. A broad **suppression dip** from TRZ near 5 mHz

2. Aether Noise Spectrum Model

The total stochastic power spectral density (PSD) in the LISA band:

$$S_{\text{UQFF}}(f) = S_{\text{GR}}(f) \hat{\alpha} [1 + P_{\text{aether}}(f)] \hat{\alpha} F_{\text{TRZ}}(f)$$

where the GR background follows $S_{\text{GR}}(f) \propto \Omega_{\text{GW}}(f) \propto f^{2/3}$ (inspiral-dominated), and:

$$S_{\text{UQFF}}(f) = S_{\text{GR}}(f) [1 + P_{\text{aether}}(f)] F_{\text{TRZ}}(f)$$

$$P_{\text{aether}}(f) = U_m \sum_n e^{-n/2} \delta(f - n f_U) W(\Delta f)$$

$$F_{\text{TRZ}}(f) = 1 - 0.1 \exp\left[-\frac{(f - f_{\text{TRZ}})^2}{2\sigma_{\text{TRZ}}^2}\right]$$

Key numerical results: $U_m = 1.0e-4$, $f_U(\text{harmonic } 1) = 9.9e-1 \text{ mHz} = 9.9e-4 \text{ Hz}$, $\Omega_{\text{GW}} \sim 1.0e-9$, F_{TRZ} peak suppression = $1.0e-1$ (10%)

$$P_{\text{aether}}(f) = U_m \tilde{I}^2(f) \exp(-n/2) \tilde{I}'(f - n f_U) W(\tilde{I}(f))$$

with spectral line width $W(\tilde{I}(f)) \approx 10\%$ fractional bandwidth, and:

$$F_{\text{TRZ}}(f) = 1 - 0.1 \exp[-(f - f_{\text{TRZ,peak}})^2 / (2\tilde{I}_{\text{TRZ}}^2)]$$

3. Spectral Components

Component	Frequency	Amplitude	Origin
GR SGWB	0.1–100 mHz	$\hat{I}_{\odot \text{GW}} \sim 10^{-10}$	Inspiral population
U_m harmonic $n=1$	0.99 mHz	$U_m \tilde{I} \sim 1.0$	Fundamental aether line
U_m harmonic $n=2$	$\sim 2 \text{ mHz}$	$U_m \tilde{I} e^{-1/2} \sim \hat{A}^1$	1st harmonic
U_m harmonic $n=3$	$\sim 3 \text{ mHz}$	$U_m \tilde{I} e^{-3/2} \sim \hat{A}^1 \hat{A} \cdot \hat{\mu}$	2nd harmonic
U_m harmonics $n=4,5$	$\sim 4, 5 \text{ mHz}$	$U_m \tilde{I} e^{-n/2} \sim \hat{\mu}^2 / \hat{A}^2$	Higher harmonics
$\hat{I}_m^2$ sidebands	$f_n \pm 0.01 \text{ mHz}$	$\hat{I}_m^2 \tilde{I}$ amplitude	Modulation pairs
TRZ suppression	$\sim 5 \text{ mHz}$	$\approx 10\%$	Trans-zero dip

4. Quantitative Results

Metric	Value
Frequency range	0.10 – 100 mHz
Spectral bins	200
Spectral resolution $\hat{I}(f)$	$8 \times 10^{-10} \text{ mHz}$
Observation time	4 years
Aether power fraction $P_{\text{aether}}/P_{\text{GR}}$	222.93%
Peak aether frequency f_{peak}	0.99 mHz
Integrated detection SNR	12,695,834
Detectable (SNR > 5)	$\approx \text{True}$

The 222.93% aether power fraction means the UQFF aether signal is more than **twice** the GR SGWB in the LISA band—an unmistakably strong signature.

5. Comparison With Known Noise Sources

Source	Spectrum Shape	Prediction	UQFF Distinguishable?
GR SGWB (inspiral)	$\propto f^{-(2/3)}$	$\hat{I}_{\odot \text{GW}} \sim 10^{-10}$	Baseline

Source	Spectrum Shape	Prediction	UQFF Distinguishable?
Galactic WD foreground	Peaked at ~ 3 mHz	Reduced by UQFF factor	Yes $\hat{=}$ WD band modified
LISA instrument noise	$\hat{=}$ $f_{\text{a}} > \hat{f}$ (low f)	Not modified	Yes $\hat{=}$ different spectral shape
Cosmological SGWB	Flat (inflation)	Not modified	Yes $\hat{=}$ UQFF adds lines
UQFF aether lines	Harmonic comb	222.93%	Unique signature

6. Validation

Run command (Windows-safe):

```
set PYTHONUTF8=1
python validate_lisa_extended.py
```

Test status from `compute_aether_noise_spectrum()`:

Check	Result
$P_{\text{aether}}/P_{\text{GR}} > 100\%$	$\hat{=}$ PASS (222.93%)
f_{peak} identified	$\hat{=}$ PASS (0.99 mHz)
Integrated SNR > 5	$\hat{=}$ PASS (12,695,834)
Detectable = True	$\hat{=}$ PASS

TEST STATUS: PASSED: `compute_aether_noise_spectrum`

7. Observational Strategy for LISA

1. **Targeted narrow-band search at 1, 2, 3 mHz:** Comb filter for U_m harmonic signature with 10% bandwidth windows
2. **Cross-correlation test:** Compare LISA data-stream with predicted sideband pattern at $f_n \hat{\pm} f_{\text{mod}}$ ($f_{\text{mod}} = 0.01$ mHz)
3. **TRZ notch identification:** Low-pass filter around 5 mHz to identify broad suppression against WD foreground
4. **Time-series modulation:** Look for $\sim 10\%$ amplitude oscillations at ~ 1 -hour period in GW background estimation

8. Conclusion

UQFF aether fields create a distinctive spectral fingerprint in the LISA band: a harmonic comb at multiples of ~ 1 mHz with 222.93% integrated power relative to the GR background, and a $\sim 10\%$ TRZ suppression near 5 mHz. With detection SNR > 12 million,

4th dissipation term parameters (PAPER_420):

$\hat{I}_{\text{c}} = 10^4 \text{ kg/m}^3$, $\hat{I}_{\text{q}} = 10^4 \text{ kg/m}^3$, $\hat{I}_{\text{H}} = 10^4 \text{ kg/m}^3$, $\hat{I}_{\text{A}} = 10^4 \text{ kg/m}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	10^4 kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{H}}$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{\text{g_sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{I}_{\text{H}}$)	Multi-scale field interactions
Buoyant	$\hat{I}_{\text{H}}^2 \hat{I}_{\text{c}} U_{\text{bi}}$	Expanding nebulae, stellar winds
Superconductive	$U_{\text{m}} \hat{I}_{\text{c}} (1 + 10^4 \hat{I}_{\text{H}}^3 \hat{I}_{\text{A}} \cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.